

## 1. Introduction

This report analyzes the errors made by our fine-tuned DeBERTa model when classifying emotions in Kitchen Nightmares dialogue. We chose this pre-trained model because it was trained on data, which has similar characteristics to reality TV - short sentences, informal language, and lots of emotion. The model was trained on 23,689 examples and tested on 1,210 examples, achieving 73.6% accuracy and F1-score of 0.734. Even though this was close to the minimum target of 0.75, there were 320 misclassifications (26.4% error rate) that show some clear patterns.

## 2. Performance Breakdown

Looking at Figure 1 (the three bar charts showing Precision, Recall, and F1-Score by emotion), performance varies a lot across the seven emotion classes:

### Best performing:

- Neutral: F1=0.789 (391 test examples)
- Happiness: F1=0.789 (278 test examples)
- Surprise: F1=0.763 (126 test examples)

### Worst performing:

- Fear: F1=0.429 (59 test examples)
- Disgust: F1=0.604 (72 test examples)
- Sadness: F1=0.635 (129 test examples)

Fear is particularly bad - with only 35.6% recall, the model misses almost 2 out of 3 fear instances. This is a problem for analyzing Kitchen Nightmares episodes because fear moments (like chefs stressed about failing businesses) won't be detected properly.

## 3. Main Error Patterns

### 3.1 Happiness vs Neutral Confusion

The biggest error pattern is confusion between happiness and neutral, which happens 71 times total (36 happiness→neutral, 35 neutral→happiness). You can see this clearly in Figure 2 (the confusion matrix) where the happiness row shows 36 instances predicted as neutral, and the neutral row shows 35 instances predicted as happiness.

Examples of these errors:

- "owner he has ever met" (true: happiness, predicted: neutral)
- "the portions are enormous" (true: happiness, predicted: surprise)

This makes sense because Kitchen Nightmares has a lot of sarcasm. Gordon Ramsay says positive-sounding things in a negative way all the time. The model only sees the text tokens, not his tone of voice or facial expressions, so it can't tell if something is actually positive or just sarcastic. The model tokenizer breaks these into neutral-looking tokens and the model struggles without the prosodic features (tone, stress) that would make the emotion clear.

### 3.2 Negative Emotions Getting Predicted as Sadness

The second major problem is that the model predicts sadness way too often when the true label is a different negative emotion. Looking at Figure 2 (confusion matrix) and Figure 3 (top confusion pairs):

- Fear → Sadness: 13 times (34.2% of fear errors)
- Disgust → Sadness: 14 times (46.7% of disgust errors)
- Anger → Sadness: 15 times (31.9% of anger errors)

Examples:

- "you're sinking" (fear → sadness)
- "it looks like a giant diaper stuffed with meatballs" (disgust → surprise)

This happens because the model learned to detect negative valence (negative vs positive emotions) but didn't learn the fine-grained differences between specific negative emotions. There are two main reasons:

1. **Class imbalance:** The training data had way more sadness examples (3,076) than fear (1,723), disgust (679), or anger (1,717). Even though we used weighted cross-entropy loss to try to fix this (with higher weights for rare classes like disgust weight=2.650 vs sadness weight=0.585), the model still saw sadness so many more times that it learned stronger associations for sadness tokens.
2. **Similar language in Kitchen Nightmares context:** In failing restaurant scenarios, fear, disgust, and sadness often use similar negative language about hopelessness and defeat. Without more context, the model can't distinguish between these subtle differences.

### 3.3 Surprise Misclassified as Neutral

Figure 3 shows "surprise → neutral" as the 3rd most common confusion (11 instances). Looking at examples like "denver's best pizza, that is a bold statement", the problem is that surprise in Kitchen Nightmares is often expressed through neutral-sounding declarative sentences instead of exclamations like "wow!" or "oh my god!". The model was pre-trained on Twitter where surprise usually has explicit markers (emojis, exclamation marks), so it doesn't transfer well to the more subtle surprise expressions in professional TV dialogue.

## 4. Sentence Length Analysis

Figure 4 shows two important visualizations about sentence length. The left panel (Word Count Distribution) shows that misclassified sentences (red bars) tend to be slightly longer than correctly classified ones (green bars). The t-test confirms this is statistically significant ( $t=-5.249$ ,  $p<0.0001$ ) - correct predictions average 5.0 words while incorrect ones average 5.9 words.

The right panel (Accuracy by Sentence Length) shows an interesting U-shaped pattern:

- 0-5 words: 79% accuracy (n=694)

- 6-10 words: 64% accuracy (n=456)
- 11-15 words: 80% accuracy (n=60)

Very short sentences perform well because they usually have strong emotion words like "disgusting!" or "perfect!" that map directly to emotions. The model's embeddings have learned these associations well. Medium sentences (6-10 words) perform worst because they add complexity without enough context - for example, "they're not crap, and they're delicious" (anger → happiness) has both negative and positive words which confuses the model. Longer sentences (11-15 words) perform better again because there's enough context for the attention mechanism to figure things out, but there aren't many long sentences in the dataset (only 60 examples).

Since most Kitchen Nightmares dialogue is short (694/1210 are  $\leq 5$  words), optimizing for longer sentences wouldn't help much anyway.

## 5. Class Imbalance Problem

There's a strong correlation between how many test examples a class has and its F1-score. Fear has only 59 test examples (4.9%) and gets  $F1=0.429$ , while neutral has 391 examples (32.3%) and gets  $F1=0.789$ . This makes sense - transformer models need lots of examples to learn good representations through their attention mechanism.

Looking at Figure 2, the fear row shows only 21 correct predictions out of 59 total (64.4% error rate), while the neutral row shows 307 correct predictions out of 391 (21.5% error rate). The weighted loss function wasn't enough to fix this because even with higher loss weights, the model just sees neutral examples 17x more often than disgust during training, so gradient updates favor the majority classes.

For the client, this means rare but important moments like fear of business failure or disgust at food quality will be missed systematically. This could mess up the emotional arc analysis of episodes.

## 6. Recommendations

Based on the patterns in Figures 1-4, here are my suggestions for improvement:

1. **Oversample minority classes:** Use SMOTE or just duplicate examples to get at least 2,000 examples per emotion. This directly addresses the imbalance shown in Figure 1.
2. **Two-stage classification:** First train a model to predict positive/negative/neutral valence, then train separate models for specific emotions within each valence group. This should help with the sadness-collapse problem in Figures 2 and 3.
3. **Add context windows:** Include the previous 2-3 utterances as input (separated by [SEP] tokens) so the model can learn from conversation flow instead of isolated sentences.
4. **Try focal loss:** This down-weights easy examples and focuses on hard ones, which might help minority classes more than simple class weighting.

5. **Ensemble with lexicon features:** Build a simple logistic regression using NRC Emotion Lexicon, then ensemble with DeBERTa. This adds explicit emotion-word mappings that might help with obvious cases like "disgusting" → disgust.

## 7. Conclusion

The DeBERTa model works reasonably well for emotion classification in Kitchen Nightmares but has clear limitations. The main issues are: (1) confusion between happiness and neutral due to sarcasm, (2) over-prediction of sadness for other negative emotions due to class imbalance, (3) poor performance on fear and disgust due to limited training data, and (4) difficulty with medium-length sentences that have conflicting emotion signals.

For the client's content analysis pipeline, the 26.4% error rate means the automated labels should be treated as noisy first-pass predictions rather than ground truth, especially for fear detection which has a 64.4% error rate. The model is functional but needs human review for production use.

## Annexure

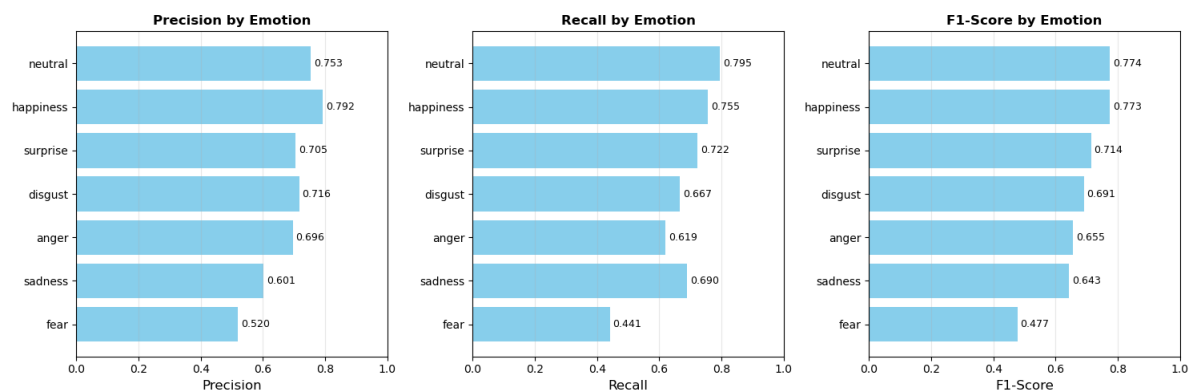


Figure 1

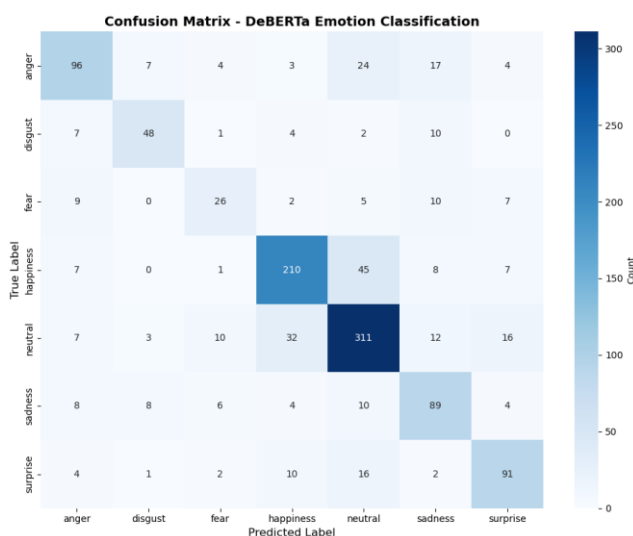


Figure 2

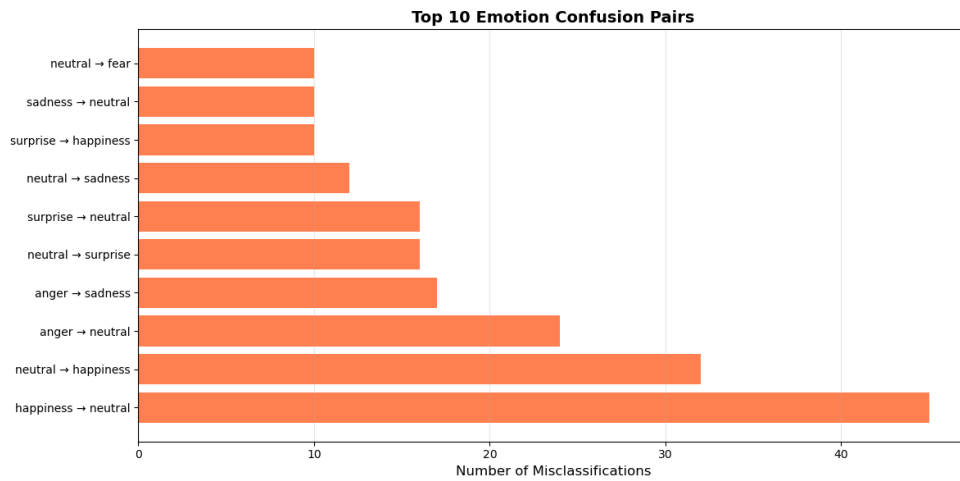


Figure 3

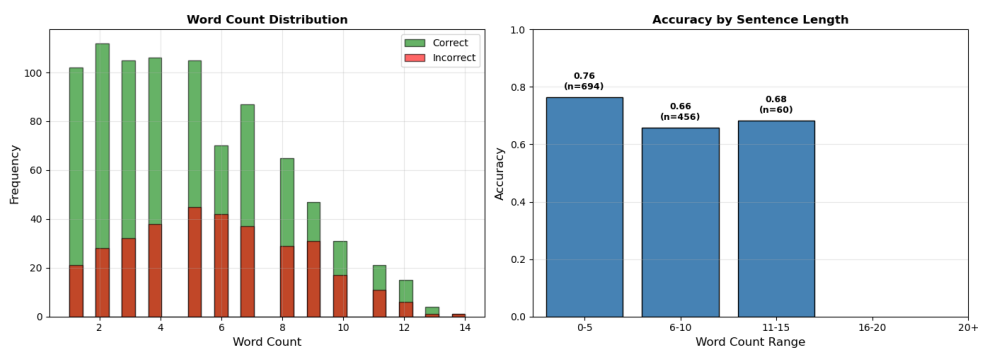


Figure 4